

Fractions and roots

A slash builds a fraction, numerator over denominator.

Parentheses decide what belongs on each side:

$$\frac{1}{2}, \quad \frac{a+b}{c+d}, \quad \frac{dy}{dx}$$

The `frac` function does the same job and is handy when a bare slash would read ambiguously; `binom` stacks two terms without a rule:

$$\frac{1}{1 + \frac{1}{x}}, \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

`sqrt` takes one argument for a square root. Give `root` an index and a radicand for anything higher:

$$\sqrt{2}, \quad \sqrt{x^2 + y^2}, \quad \sqrt[3]{x}, \quad \sqrt[n]{x}$$